

Systemic Risk vs. Systematic Risk: What's the Difference?

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Here at *Investopedia*, we emphasize the importance of prudent investing—put at stake only what you can afford to lose and ensure your choices align with your financial goals and risk tolerance. This doesn't just mean preparing for "specific" or "unsystematic" risks, such as the potential bankruptcy of a company whose stock is in your portfolio or interest rate changes affecting your bonds. Yuehua Tang, Ph.D., a researcher and professor of finance at the University of Florida, says that individual investors and large firms need to be prepared for more than that.¹ Tang, who has been investigating how companies manage the risks of climate change, says we need to be ready for both systemic, the risk of one part of a market setting off a major collapse, and systematic risks, which are widespread.² Tang says there are ways to mitigate these risks, and we detail some below.

Key Takeaways

- Systemic risk and systematic risk are both dangers to the financial markets and economy, but the cause of these risks and the approaches for managing them are different.
- Systemic risk is the risk that a company or industry-level risk could trigger a major collapse.
- Systematic risk is the risk inherent to the entire market, which can be economic, sociopolitical, technological, or natural in origin.
- Systemic risk is harder to quantify and harder to predict, while systematic risk is more quantifiable and predictable.

While specific risks can be reduced through diversification, systemic and systematic risks pose broader threats that are more formidable because of their widespread impact on the financial system and the market as a whole. Systemic risk is when a single company or sector failure could trigger an economic crisis, like a match that, once lit, threatens to burn a whole building down.³ By contrast, systematic risk involves pervasive threats like economic recessions, geopolitical unrest, or natural disasters—akin to the ever-present possibility of a storm knocking your house down. This risk doesn't appear one day when someone brings home matches but is always there, no matter what you do.

Systemic and systematic risks have different sources and thus require their own approaches to managing them. Tang suggests that asset allocation, staying informed, and hedging strategies are vital in navigating these risks—approaches we discuss below. He uses the example of climate risk mitigation in Florida, where he teaches. "You might

worry about how your home will be affected by climate change," such as by flooding. "That will affect the value of your home. But if you're proactive, you can raise the foundation and might not be affected."

Unsystematic or Specific Risk

Before turning to systemic and systematic risks and how they can be managed or prepared for, it's perhaps best to turn to typical forms of portfolio risks and how investors anticipate and mitigate them to see how the strategies mentioned later are different.

Unsystematic or specific risk is unique to a particular company or industry and can be mitigated by diversifying and holding a broad portfolio of investments. Unlike systematic risk, which affects the entire market or a large segment of it, specific or unsystematic risk relates to a single company, sector, or industry. And unlike systemic risk, these generally do not spark problems in the financial system as a whole. Here are two types:

- **Company-specific risk:** This could involve the risk of investing in a company that ends up performing poorly because of management decisions, financial instability, product recalls, or legal challenges that affect its profitability and stock price.
- **Industry-specific risk:** This involves risks in a particular industry. For example, technological changes could make products or services in one sector obsolete, while regulatory changes could significantly affect the profitability of companies in another industry.

It's easy to imagine other specific risks for investments in currencies, bonds, commodities, and other assets. By diversifying your portfolio across different companies, industries, and asset types, you can cut the overall risk for your portfolio. This is because the strong performance of others can offset the underwhelming performance of one investment.

Systemic Risk

Systemic risk is connected to the complete failure of a business, a sector, an industry, a financial institution, or the overall economy. The critical part, as the editors of the *Handbook of Systemic Risk* note, is that "the risk that the ability of the system to function as intended is seriously degraded. It does not refer to the risk that a component or several components fail unless such failure jeopardizes the integrity of the system," with this last part separating it from the specific risks covered earlier.³

Systemic risk typically involves a cascading set of dangers that multiply and could soon engulf a sector or economy because of how the modern economy is so interconnected, not just in this or that region or country, but across the world. Mitigating the risk of the initial spark is easier than trying to control what happens once the disaster gets rolling.

The collapse of Lehman Brothers in September 2008 is a textbook example of a cascading risk that led to a global financial crisis. This event illustrates how systemic risk can trigger adverse effects throughout the financial system and the broader economy.

1. **A company goes under:** Lehman Brothers filed for Chapter 11 bankruptcy protection Sept. 15, 2008, the largest bankruptcy filing in U.S. history at the time. In the abstract, one might think this is a terrible blow to the employees who worked there, the company's investors, and perhaps some companies that relied on them, but no more.
2. **Panic:** However, this was amid the financial crisis, and the event indicated that the economic crisis was hitting even the most stable-seeming firms on Wall Street. The bankruptcy led to immediate panic in global financial markets as investors lost confidence in the stability of financial institutions.
3. **The panic spreads:** There was a steep decline in stock markets worldwide, spreading fears that other financial institutions might follow Lehman Brothers into bankruptcy court.
4. **The contagion brings down more firms:** Firms that had exposure to Lehman Brothers through loans, investments, or derivatives face significant losses, and investors are pulling their money out as fast as they can. The Reserve Primary Fund, for example, had \$785 million in short-term loans for Lehman Brothers, known as commercial paper, that became worthless when Lehman filed for bankruptcy. Though this was only a small portion of its assets, the company soon couldn't meet its redemption requests and was forced to liquidate.
5. **Market liquidity freezes:** Banks became wary of lending to each other because of fears of insolvency, leading to a significant tightening of credit. The credit freeze affected businesses and consumers alike, making securing loans for business operations or personal much harder to get, immediately cutting economic demand. Now, the entire economy was at risk.
6. **Governments and central banks rush in:** As the crisis veers out of control, governments and central banks worldwide launch unprecedented interventions. These included bailouts of financial institutions, liquidity support for banks and other emergency economic measures to stabilize the financial system and restore confidence.
7. **Global economic recession:** The cascading effects of Lehman's collapse contributed to a severe international economic recession. Economic growth contracted, unemployment rates soared, and many countries faced significant fiscal challenges for years. The recession was the deepest and most widespread economic downturn since the Great Depression. This means careers people never started and years of struggles for individuals and families waiting for jobs to open up and the economy to turn.

In the aftermath of the crisis, there was a global push for regulatory reforms in the financial sector. This led to stricter financial regulations and oversight to prevent a similar crisis in the future, including measures aimed at increasing financial institutions' resilience and improving financial markets' transparency, all meant to stop the initial spark that would start the cascade again.

The Paradox of Systemic Risk

What makes systemic risk such a difficulty is that, as the editors of the *Handbook of System Risk* write, "actions to reduce the default probability of each individual systemically important institution can have the result of increasing the probability of systemwide failure."³ This means that attempts to reduce problems in individual institutions or sectors in the aggregate could make a crisis more, not less, likely in certain instances.

For example, in forest fire management, it's typical to clear underbrush and use controlled smaller burns to do away with the "fuel" that could lead to massive wildfires. But by doing so, smaller fires don't happen and clearer underbrush could cause denser forests that make the area more, not less, susceptible to severe wildfires. As such, regulators and market participants need to manage these risks carefully, giving rise to debates over if and when preventative measures could be making wider problems more or less likely.

Systematic Risk

Systematic or market risk refers to the inherent danger present throughout the entire market that cannot be mitigated by diversifying your portfolio. Broad market risks include recessions, periods of economic weakness, wars, rising or stagnating interest rates, fluctuations in currencies or commodity prices, and other "big-picture" issues like climate change. Systematic risk is embedded in the market's overall performance and cannot be eliminated simply by diversifying assets. While it cannot be eradicated, systematic risk can be managed by adopting a strategic asset allocation approach.⁴

Meanwhile, unsystematic or idiosyncratic risks are hazards specific to a firm or industry and can be reduced or eliminated through diversification. Unlike systematic risk, unsystematic risk can be mitigated because it is not tied to the broader market's movements. Diversifying a portfolio across various sectors and asset classes can protect against the negative impact of these specific risks.

Here are several commonly mentioned types of systematic risk:

- **Market risk:** This type of systematic risk generally impacts all securities in the market. Changes in interest rates, inflation, and gross domestic product growth cause it. For instance, if interest rates rise, the bond value generally falls, affecting all bondholders in the market.
- **Interest rate risk:** This form of systematic risk relates to changes in the market interest rates. Generally, when interest tends to rise (for instance, because of central banks' decisions), the prices of existing fixed-income securities fall. This is because newly issued securities will have higher yields, making existing lower-yielding securities less attractive.
- **Inflation risk:** This systematic risk arises from the potential loss of purchasing power due to inflation. When inflation increases, the real return on investments decreases. This impacts all investors as the cost of goods and services increases, affecting the overall economy.

Another systematic risk often mentioned is climate change, which will affect economies and markets, policies, operational costs, real estate, commodity prices, and far more worldwide. To address the challenges of systematic risk, e.g., with climate looming, investors might incorporate a variety of asset classes into their portfolios, including equities, fixed income, cash, and real estate, because each of these will react differently to a major systematic change.

By including assets that may benefit from the transition to a greener economy, such as renewable energy companies or green bonds, investors can also align their portfolios with a long-term view of market shifts driven by environmental factors. This approach not only aids in managing systematic risk but also could contribute to a more sustainable investment strategy. Let's look at this more closely.

Example of Hedging Against Systematic Risks

Tang suggests hedging for individuals and companies to mitigate systematic risks. Here's one way this would work.

The first step would be to identify the risks posed by climate change that would impact you or your firm. These can include direct physical risks from climate-related natural disasters (like floods, hurricanes, and wildfires), transitional risks from changing regulations (such as carbon taxes or emissions trading schemes), and market risks from shifts in consumer preferences toward more sustainable products and services.

After identifying the risks, the next step is to assess what's at risk. What in your operations or investment portfolio are most affected by shifts in climate? Based on the risk assessment, you can then develop a hedging strategy. Here are some possible moves:

- **Diversification:** Investing in a range of assets or businesses is less likely to be affected by the same climate events.
- **Derivatives:** Using financial derivatives like options, futures, and swaps to hedge against specific risks. For example, an agricultural company might use futures contracts to lock in prices for crops that could be affected by extreme weather.
- **Insurance:** Buy policies that specifically cover climate-related damage or business interruptions.
- **Investing in green technologies:** Allocate capital toward renewable energy, energy efficiency, or other green technologies that could benefit from transitioning to a low-carbon economy.

Once you choose your hedging strategies, make your strategic investments. You'll need to continuously monitor and adjust to changes in the market and climate since shifts in both can and will happen.

Systemic Risk vs. Systematic Risk Examples

The collapse of Lehman Brothers Holdings Inc. in 2008 is an example of systemic risk. After the global financial services firm filed for bankruptcy, shock waves were felt throughout the entire financial system and the economy. Because Lehman Brothers was a large company and deeply ingrained within the financial system, its collapse resulted in a domino effect that created a major risk to the global financial system.⁵

Meanwhile, the Great Recession can be viewed through the lens of systematic risk, demonstrating how broad economic downturns impact the entire market and all types of investments. During the recession, the market saw a significant reshuffling of asset values: riskier securities had a sell-off, while traditionally safer assets, like U.S. Treasury securities, increased in value.

This period vividly illustrates the nature of systematic risk, where the entire financial market is subject to the effects of widespread economic challenges, affecting investments across the board.⁶ But it also shows that there can be relationships between systematic and systemic risks, where an economic slowdown causes problems in a firm or industry, which then spark panic, spreading out to threaten the global economy.

Key Differences Between Systemic Risk and Systematic Risk

Systemic Risk

- This term refers to the risk of collapse(s) of an entire financial system or market.
- Triggered by the interconnectedness of institutions, like a domino effect.
- Usually caused by unforeseen events or financial crises.

Systematic Risk

- This term represents the overall risk inherent in the entire market.
- Affects all securities in the same manner, because of macroeconomic factors.
- Examples include inflation rates, interest changes, and political instability.

Can We Eliminate Systematic Risk?

Systematic risk cannot be eliminated through simple diversification because it affects the entire market, but it can be managed to some effect through hedging strategies.

Can Systemic Risk Impact Individual Investments?

Yes, systemic risk can significantly impact individual investments as it influences the entire financial system, potentially leading to the devaluation of investments or even total loss.

Is Systematic Risk Possible To Predict?

While it's impossible to predict specific events that lead to systematic risk, investors can use indicators such as economic data, market trends, and financial models to estimate or anticipate potential market risks.

The Bottom Line

Systemic and systematic risks pose significant threats and potential challenges to the financial markets and economies around the globe. Systemic risk often stems from a company or industry-level event that could spark a broad collapse. Conversely, systematic risk is inherent to the entire market, influenced by various economic, sociopolitical, and market-related factors. Investors (private or institutional) should prepare as much as possible for both.